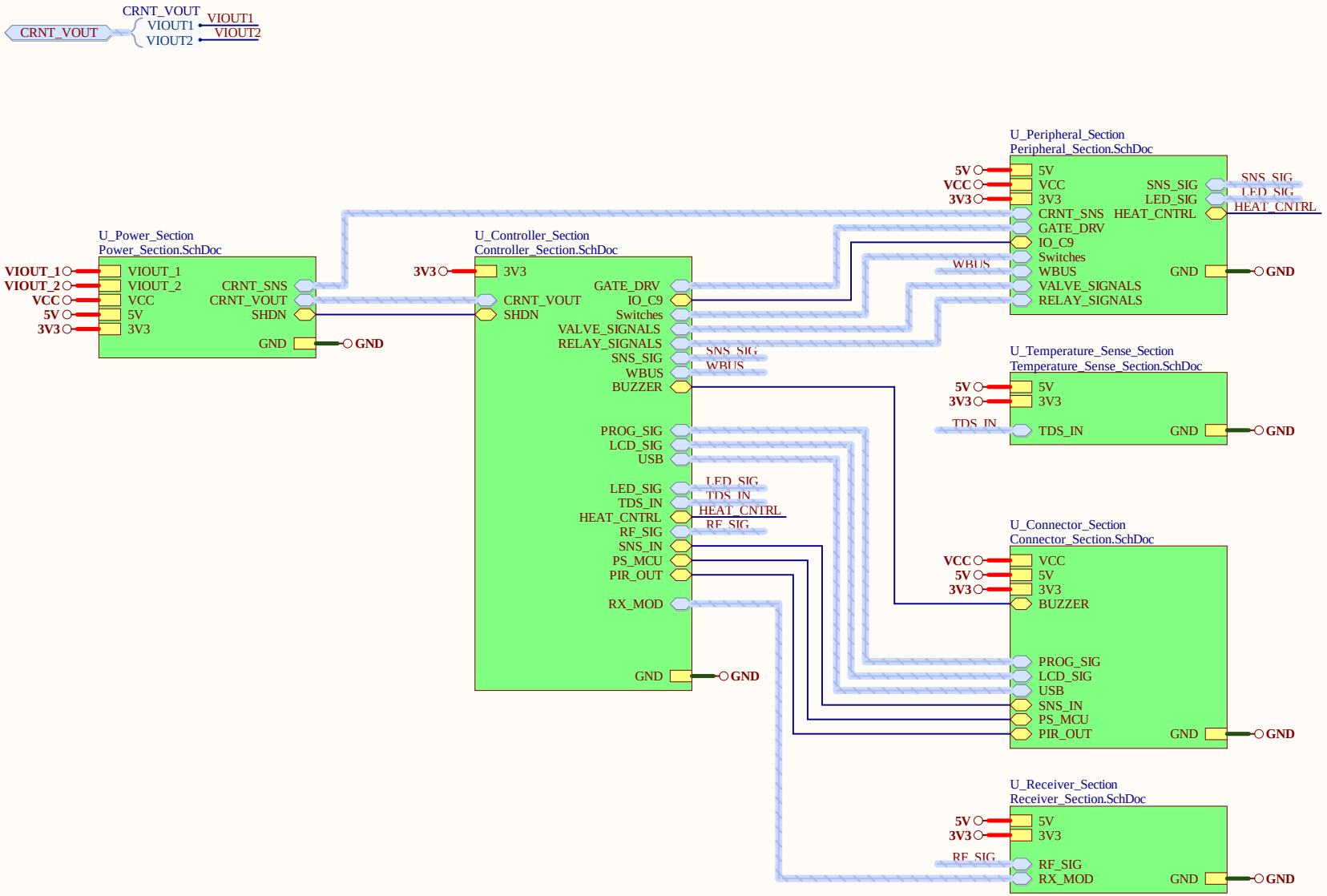
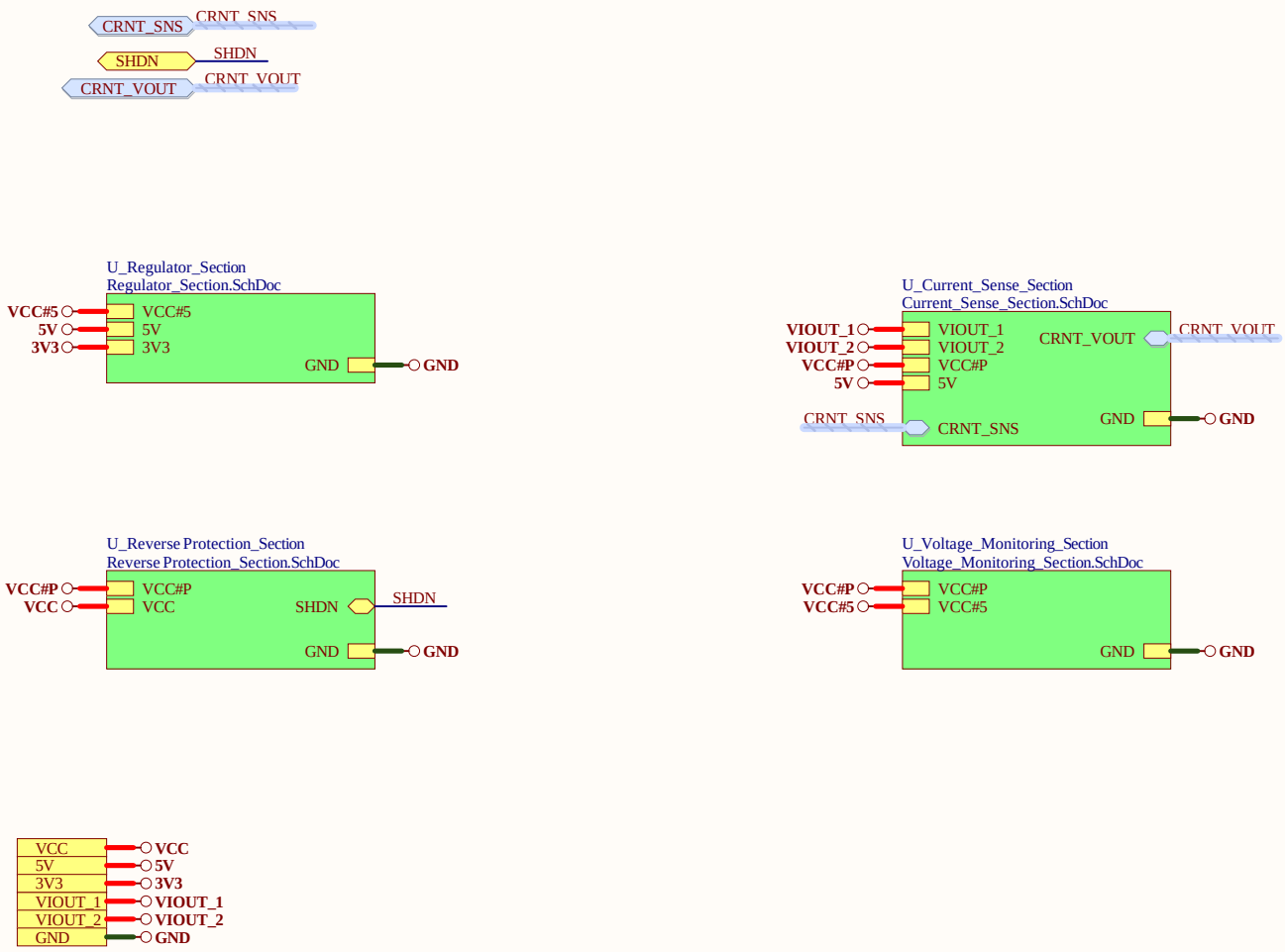
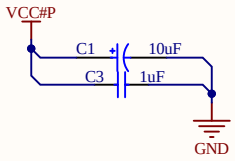

CRNT_VOUT
CRNT_VOUT
VIOUT1
VIOUT2
VIOUT1
VIOUT2
U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VIOUT_1
VIOUT_2
VCC
5V
3V3
CRNT_SNS
CRNT_VOUT
SHDN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
CRNT_VOUT
SHDN
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
RELAY_SIGNALS
SNS_SIG
WBUS
BUZZER
PROG_SIG
LCD_SIG
USB
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
SNS_IN
PS_MCU
PIR_OUT
RX_MOD
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
5V
VCC
3V3
5V
VCC
3V3
CRNT_SNS
HEAT_CNTRL
GATE_DRV
IO_C9
Switches
WBUS
VALVE_SIGNALS
RELAY_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
GND
GND
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
5V
3V3
5V
3V3
TDS_IN
TDS_IN
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VCC
5V
3V3
VCC
5V
3V3
BUZZER
PROG_SIG
LCD_SIG
USB
SNS_IN
PS_MCU
PIR_OUT
GND
GND
U_Receiver_Section
Receiver_Section.SchDoc
5V
3V3
5V
3V3
RF_SIG
RF_SIG
RX_MOD
GND
GND

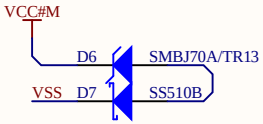

CRNT_SNS
CRNT_SNS
SHDN
SHDN
CRNT_VOUT
CRNT_VOUT
U_Regulator_Section
Regulator_Section.SchDoc
VCC#5
5V
3V3
VCC#5
5V
3V3
GND
GND
U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_VOUT
CRNT_VOUT
CRNT_SNS
CRNT_SNS
GND
GND
U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC
VCC#P
VCC
SHDN
SHDN
GND
GND
U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC#5
VCC#P
VCC#5
GND
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND



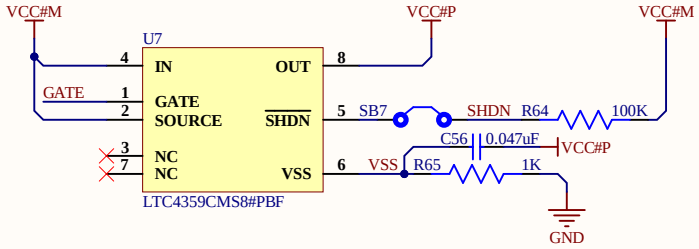
DECOUPLING CAPACITOR



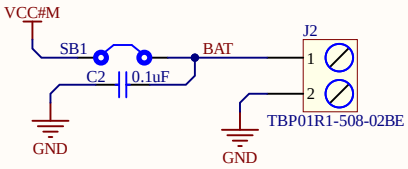
INPUT DIODES



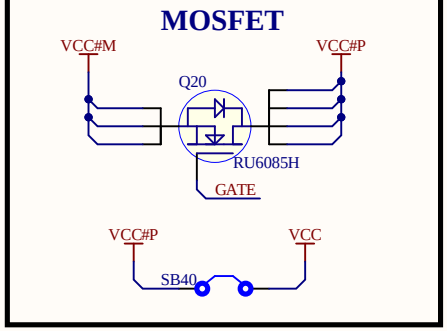
REVERSE POLARITY PROTECTION IC

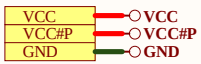


BATTERY / MAIN INPUT

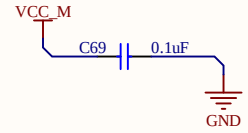


MOSFET

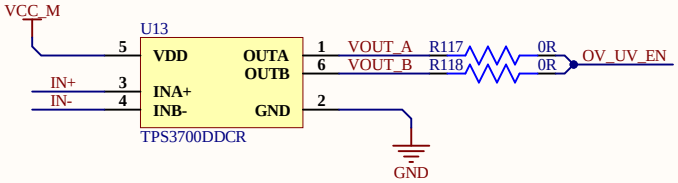




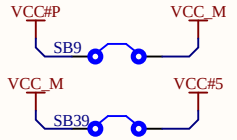
DECOUPLING CAPACITOR



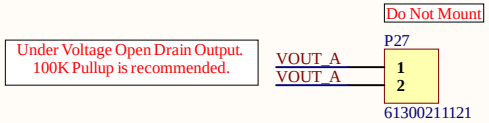
UNDER AND OVER VOLTAGE PROTECTION IC



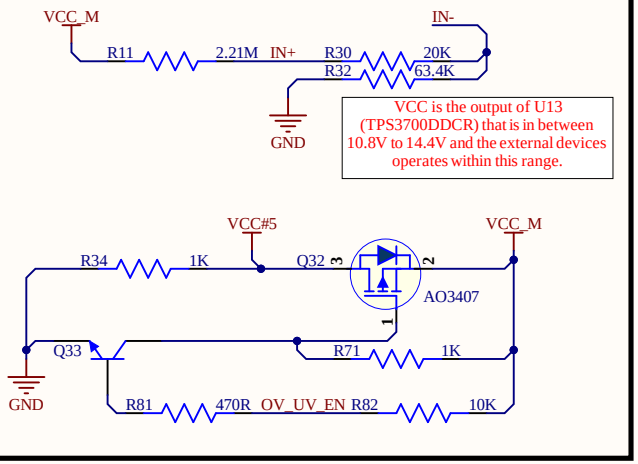
VOLTAGE TO MONITOR



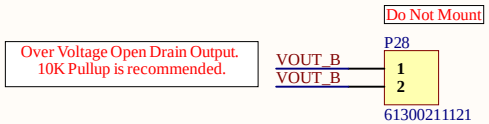
UNDER VOLTAGE OUTPUT

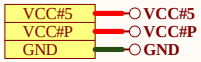


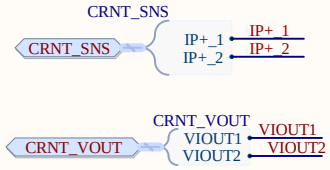
SUPPORTING CIRCUITRY

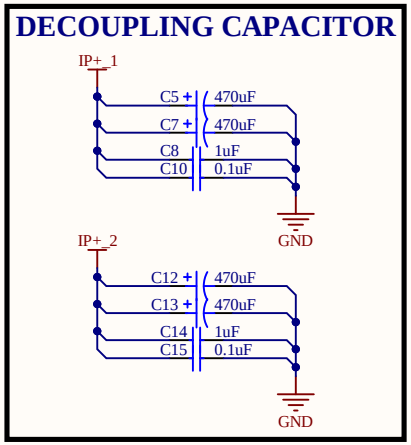


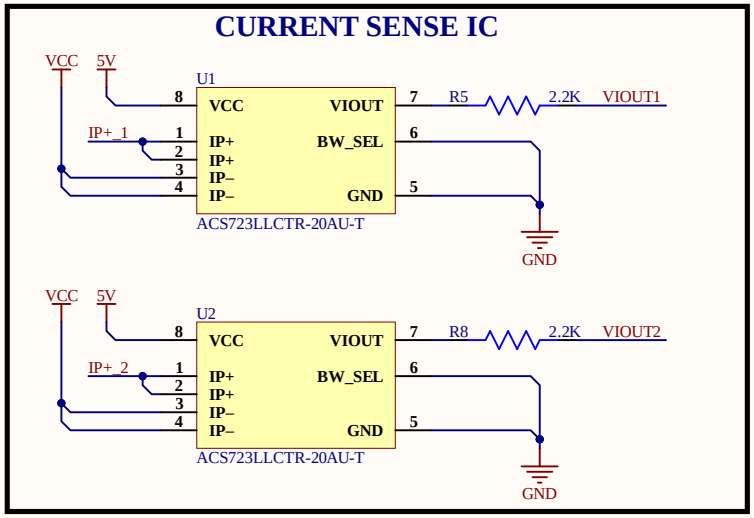
OVER VOLTAGE OUTPUT

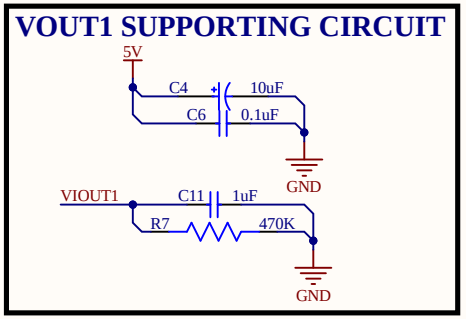


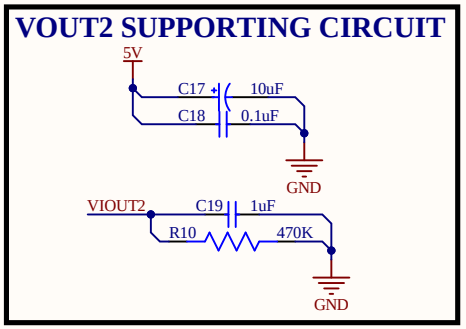




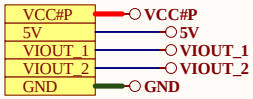


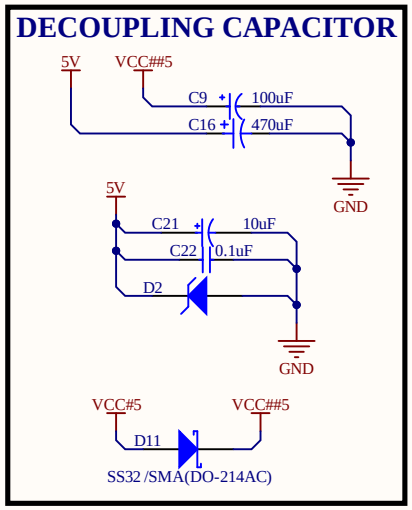


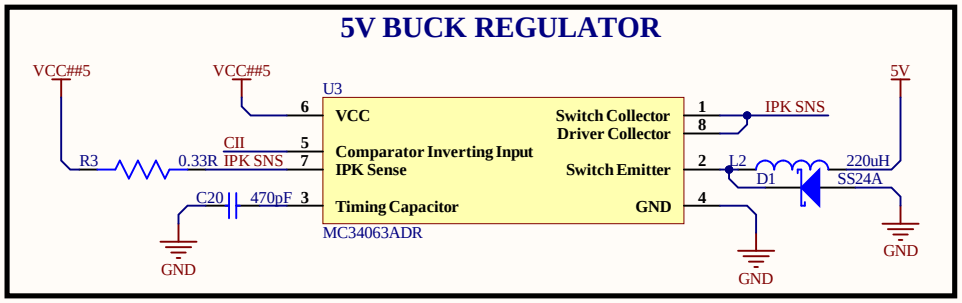


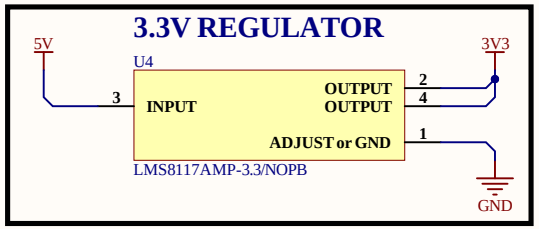


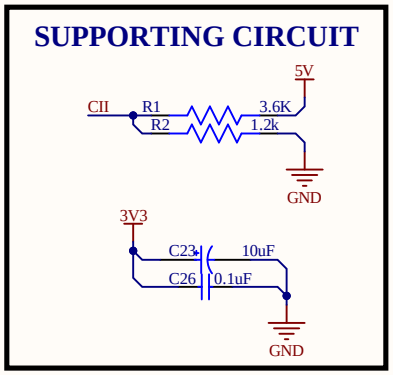
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.



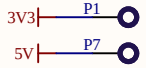


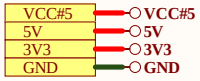


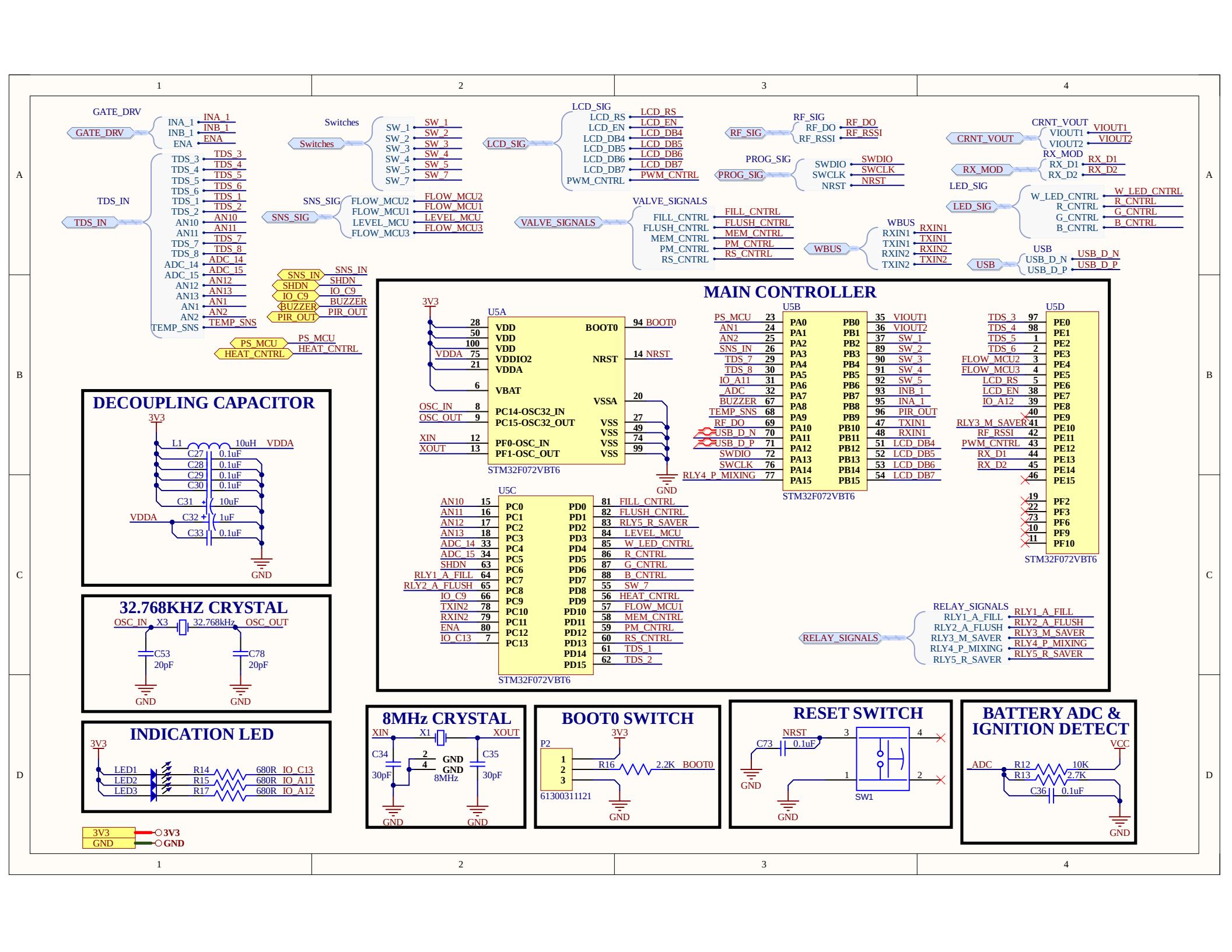


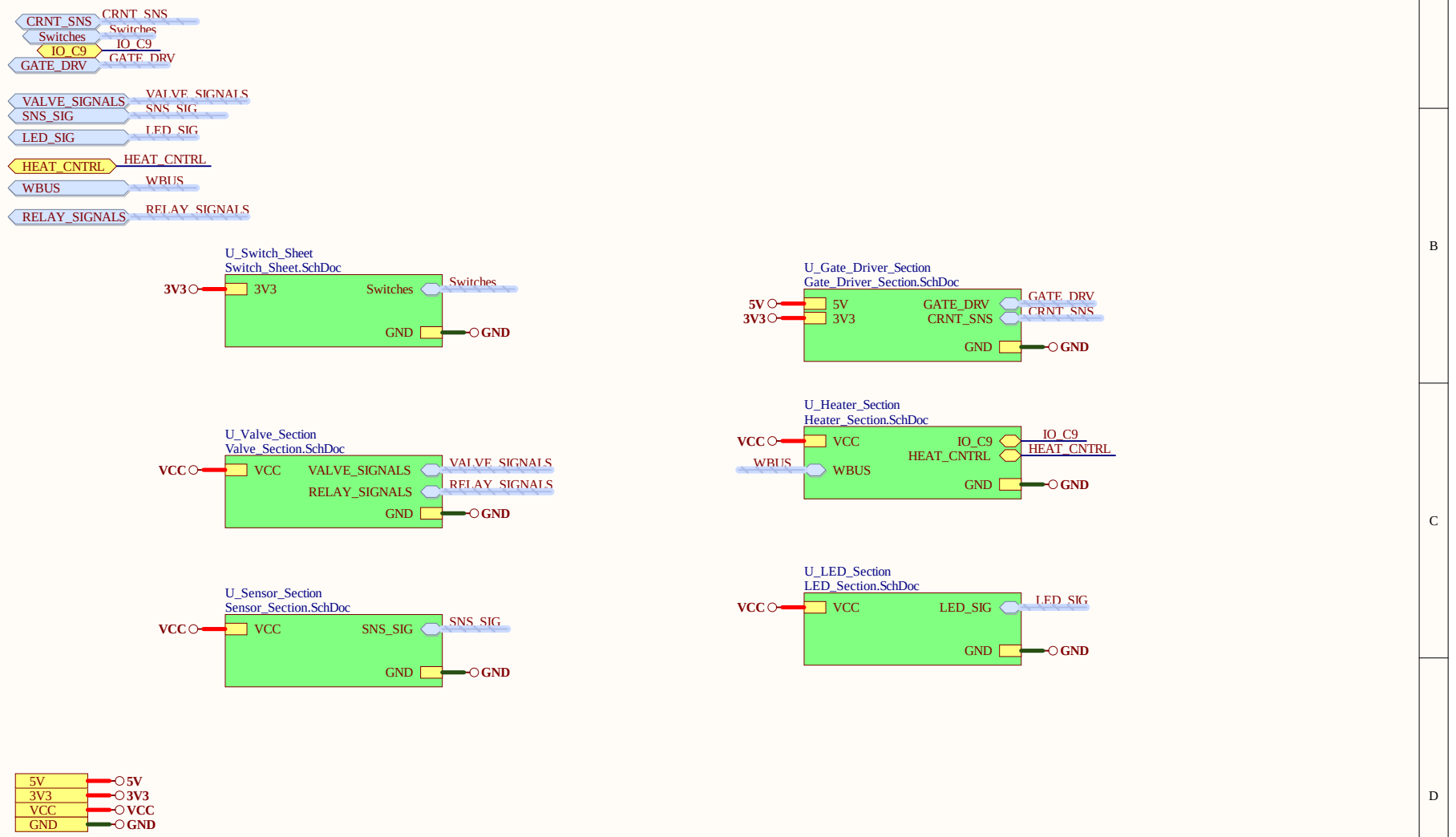


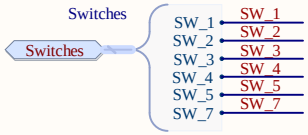
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

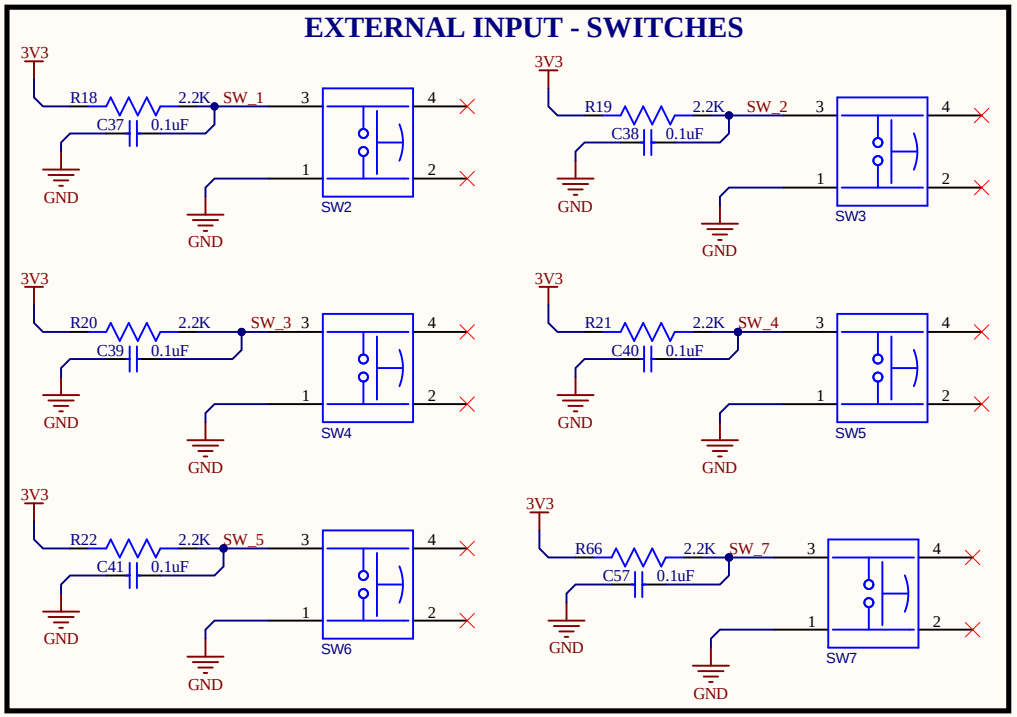




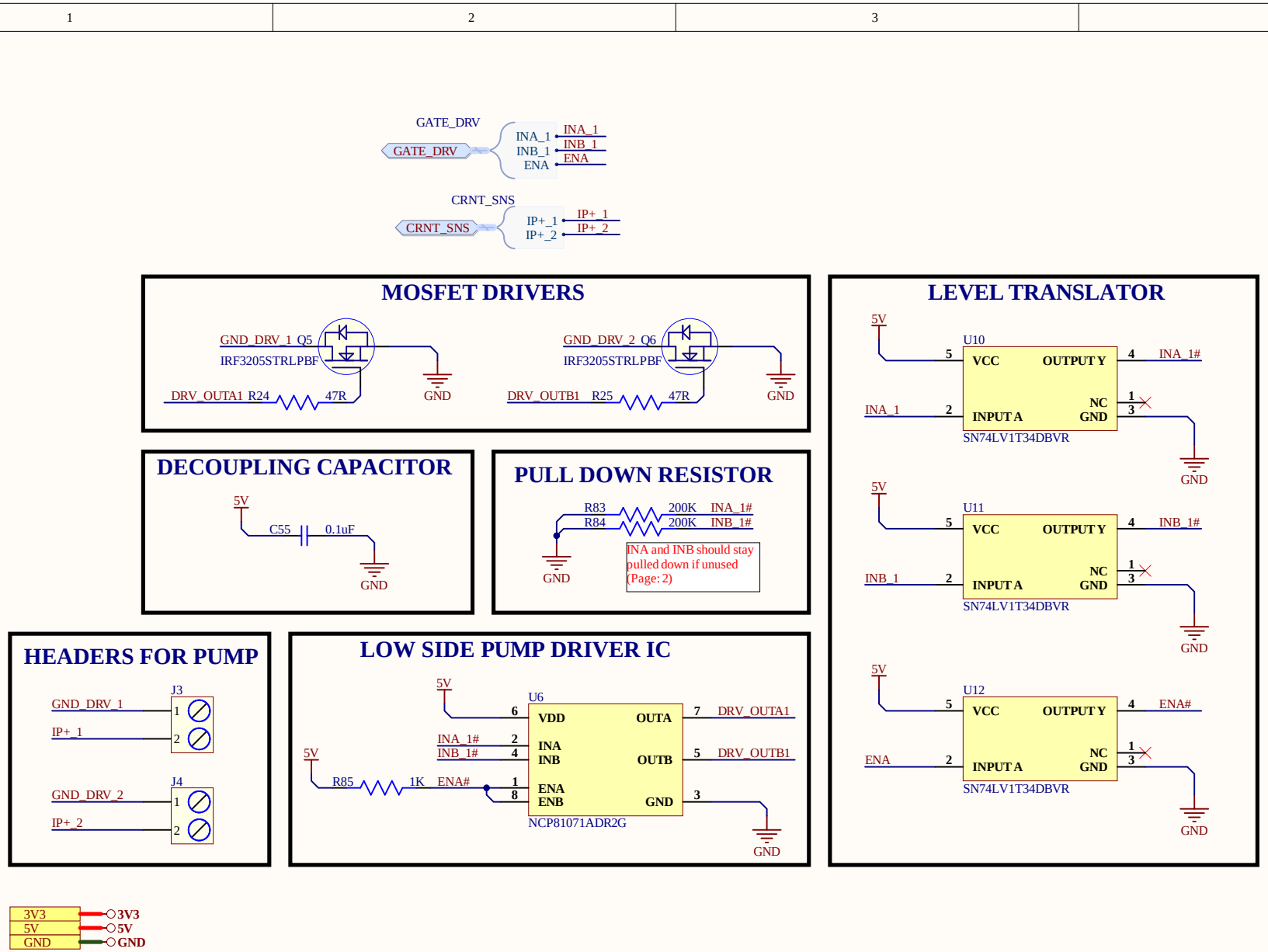

GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SNS_SIG
FLOW_MCU2
FLOW_MCU1
LEVEL_MCU
FLOW_MCU3
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
RF_SIG
RF_DO
RF_RSSI
PROG_SIG
SWDIO
SWCLK
NRST
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
USB
USB_D_N
USB_D_P
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
MAIN CONTROLLER
U5A
VDD
VDDA
VDDIO2
VBAT
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
BOOT0
NRST
VSSA
VSS
OSC_IN
OSC_OUT
XIN
XOUT
STM32F072VBT6
U5B
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
U5C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
U5D
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PF2
PF3
PF6
PF9
PF10
RELAY_SIGNALS
RLY1_A_FILL
RLY2_A_FLUSH
RLY3_M_SAVER
RLY4_P_MIXING
RLY5_R_SAVER
DECOUPLING CAPACITOR
L1
10uH
C27
C28
C29
C30
C31
C32
C33
0.1uF
1uF
32.768KHZ CRYSTAL
X3
32.768kHz
C53
C78
20pF
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R
8MHz CRYSTAL
C34
C35
30pF
8MHz
BOOT0 SWITCH
P2
R16
2.2K
61300311121
RESET SWITCH
C73
SW1
BATTERY ADC & IGNITION DETECT
R12
R13
10K
2.7K
C36
VCC
3V3
GND

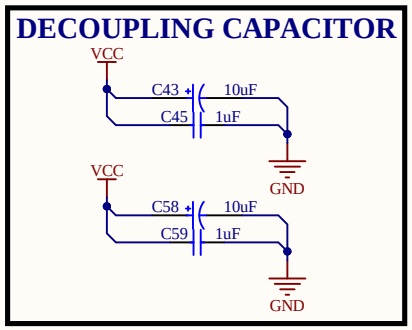

CRNT_SNS
Switches
IO_C9
GATE_DRV
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
WBUS
RELAY_SIGNALS
U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
Switches
GND
U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
3V3
GATE_DRV
CRNT_SNS
GND
U_Valve_Section
Valve_Section.SchDoc
VCC
VALVE_SIGNALS
RELAY_SIGNALS
GND
U_Heater_Section
Heater_Section.SchDoc
VCC
IO_C9
HEAT_CNTRL
WBUS
GND
U_Sensor_Section
Sensor_Section.SchDoc
VCC
SNS_SIG
GND
U_LED_Section
LED_Section.SchDoc
VCC
LED_SIG
GND
5V
3V3
VCC
GND

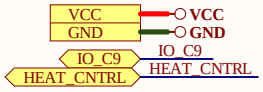

Switches
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7

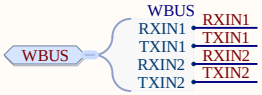

EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
SW2
R19
SW_2
C38
SW3
R20
SW_3
C39
SW4
R21
SW_4
C40
SW5
R22
SW_5
C41
SW6
R66
SW_7
C57
SW7

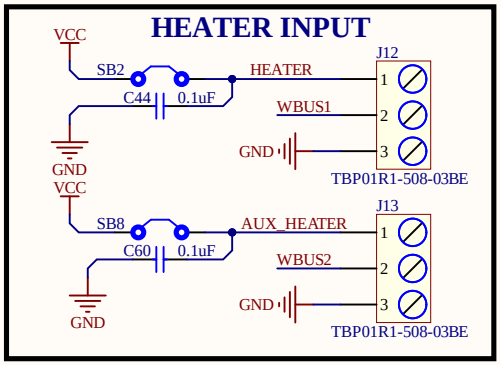

3V3
GND
3V3
GND

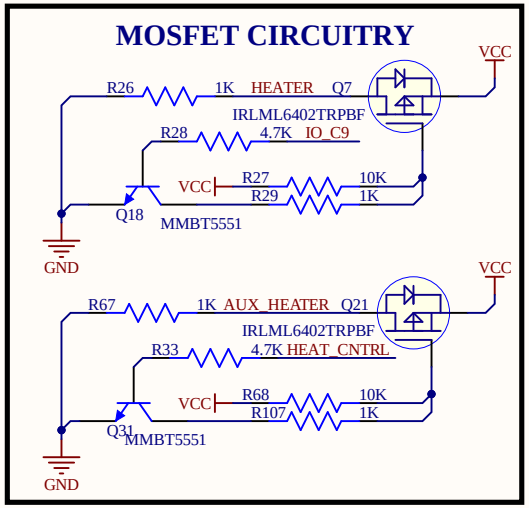

GATE_DRV
GATE_DRV
INA_1
INB_1
ENA
INA_1
INB_1
ENA
CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2
MOSFET DRIVERS
GND_DRV_1 Q5
IRF3205STRLPBF
DRV_OUTA1 R24
47R
GND
GND_DRV_2 Q6
IRF3205STRLPBF
DRV_OUTB1 R25
47R
GND
LEVEL TRANSLATOR
5V
U10
5
VCC
OUTPUT Y
4
INA_1#
INA_1
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
5V
U11
5
VCC
OUTPUT Y
4
INB_1#
INB_1
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
5V
U12
5
VCC
OUTPUT Y
4
ENA#
ENA
2
INPUT A
NC
GND
1
3
SN74LV1T34DBVR
GND
DECOUPLING CAPACITOR
5V
C55
0.1uF
GND
PULL DOWN RESISTOR
R83
200K
INA_1#
R84
200K
INB_1#
GND
INA and INB should stay pulled down if unused (Page: 2)
HEADERS FOR PUMP
J3
GND_DRV_1
1
IP+_1
2
J4
GND_DRV_2
1
IP+_2
2
LOW SIDE PUMP DRIVER IC
5V
U6
6
VDD
OUTA
7
DRV_OUTA1
INA_1#
2
INA
INB_1#
4
INB
OUTB
5
DRV_OUTB1
5V
R85
1K
ENA#
1
ENA
8
ENB
GND
3
NCP81071ADR2G
GND
3V3
5V
GND
3V3
5V
GND

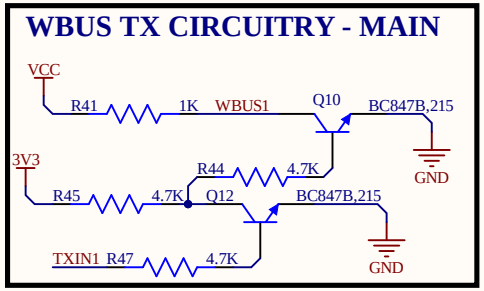

DECOUPLING CAPACITOR
VCC
C43
10uF
C45
1uF
GND
VCC
C58
10uF
C59
1uF
GND

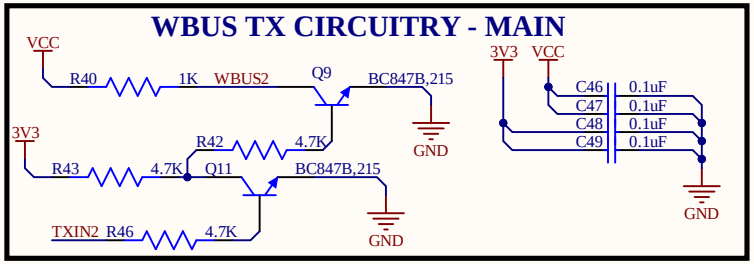

VCC
GND
IO_C9
HEAT_CNTRL
VCC
GND
IO_C9
HEAT_CNTRL

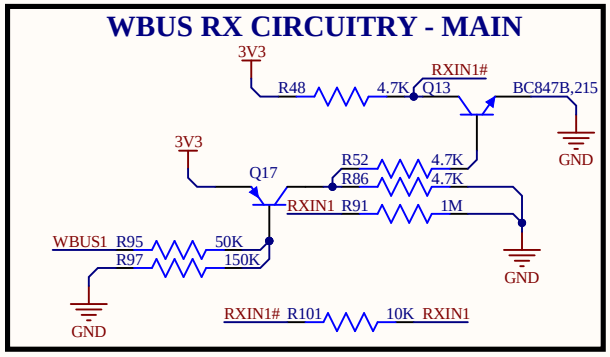

WBUS
RXIN1
TXIN1
RXIN2
TXIN2
WBUS
RXIN1
TXIN1
RXIN2
TXIN2

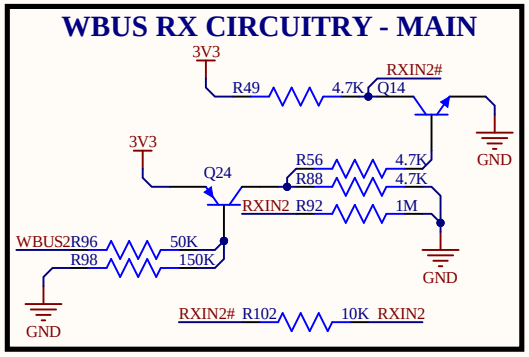

HEATER INPUT
VCC
SB2
HEATER
J12
C44
0.1uF
WBUS1
GND
GND
TBP01R1-508-03BE
VCC
SB8
AUX_HEATER
J13
C60
0.1uF
WBUS2
GND
GND
TBP01R1-508-03BE

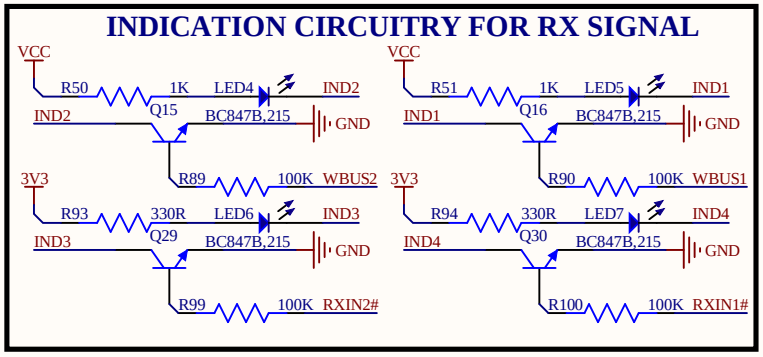

MOSFET CIRCUITRY
VCC
R26
1K
HEATER
Q7
IRLML6402TRPBF
R28
4.7K
IO_C9
VCC
R27
10K
R29
1K
Q18
MMBT5551
GND
VCC
R67
1K
AUX_HEATER
Q21
IRLML6402TRPBF
R33
4.7K
HEAT_CNTRL
VCC
R68
10K
R107
1K
Q31
MMBT5551
GND


WBUS TX CIRCUITRY - MAIN
VCC
R41
1K
WBUS1
Q10
BC847B,215
GND
3V3
R44
4.7K
R45
4.7K
Q12
BC847B,215
GND
TXIN1
R47
4.7K
GND

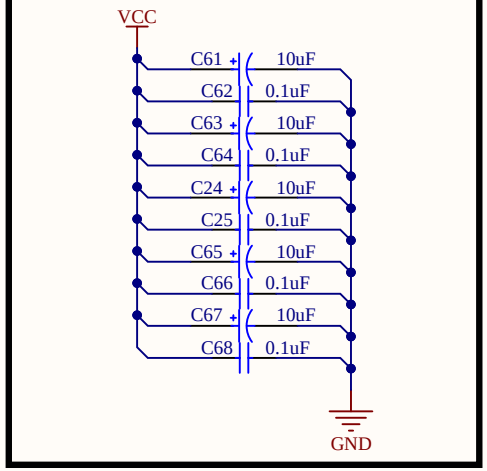

WBUS TX CIRCUITRY - MAIN
VCC
R40
1K
WBUS2
Q9
BC847B,215
GND
3V3
VCC
C46
0.1uF
C47
0.1uF
C48
0.1uF
C49
0.1uF
GND
3V3
R42
4.7K
R43
4.7K
Q11
BC847B,215
GND
TXIN2
R46
4.7K
GND


WBUS RX CIRCUITRY - MAIN
3V3
RXIN1#
R48
4.7K
Q13
BC847B,215
GND
3V3
Q17
R52
4.7K
R86
4.7K
RXIN1
R91
1M
GND
WBUS1
R95
50K
R97
150K
GND
RXIN1#
R101
10K
RXIN1

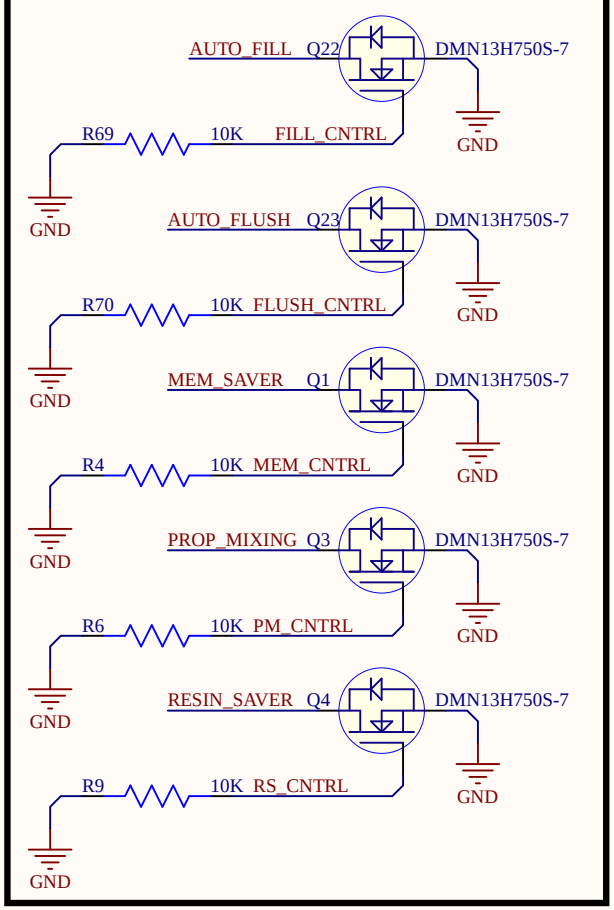

WBUS RX CIRCUITRY - MAIN
3V3
RXIN2#
R49
4.7K
Q14
GND
3V3
Q24
R56
4.7K
R88
4.7K
RXIN2
R92
1M
GND
WBUS2
R96
50K
R98
150K
GND
RXIN2#
R102
10K
RXIN2


INDICATION CIRCUITRY FOR RX SIGNAL
VCC
R50
1K
LED4
IND2
IND2
Q15
BC847B,215
GND
VCC
R51
1K
LED5
IND1
IND1
Q16
BC847B,215
GND
3V3
R89
100K
WBUS2
3V3
R90
100K
WBUS1
R93
330R
LED6
IND3
IND3
Q29
BC847B,215
GND
R94
330R
LED7
IND4
IND4
Q30
BC847B,215
GND
R99
100K
RXIN2#
R100
100K
RXIN1#

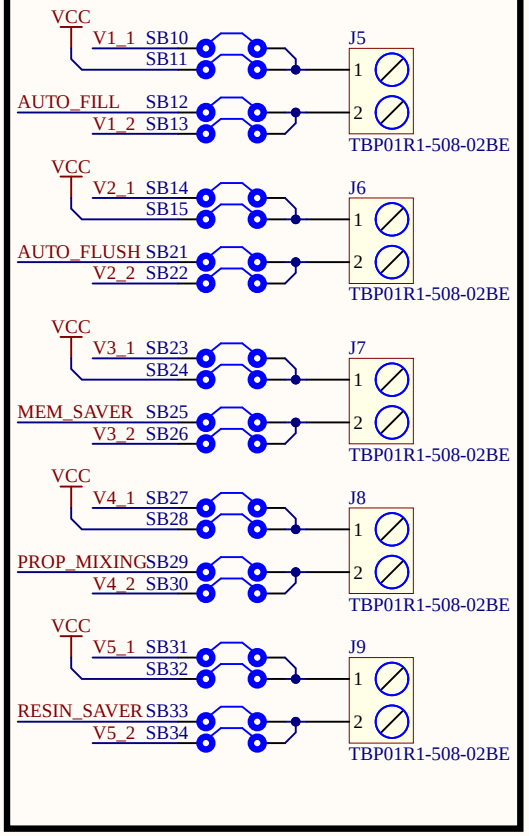
DECOUPLING CAPACITOR

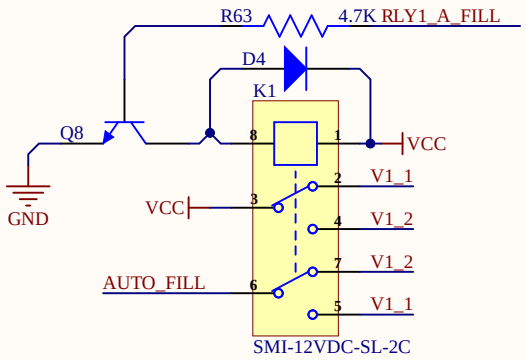


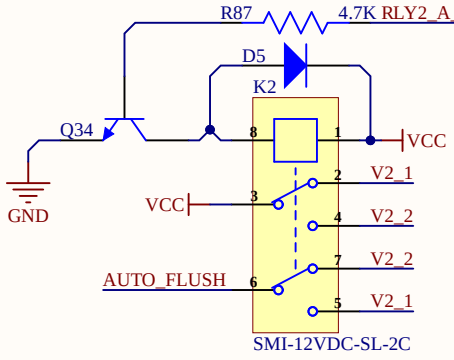
SOLENOID CONTROL CIRCUIT

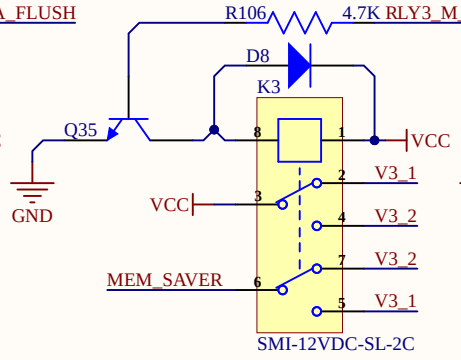


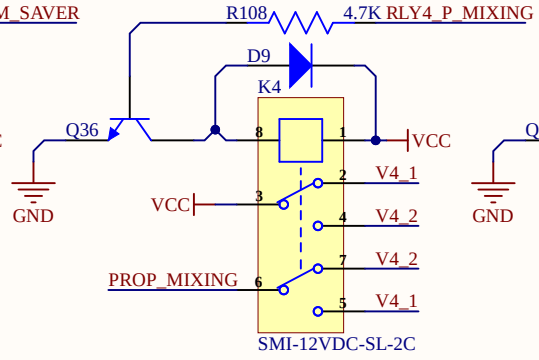
SOLENOID VALVE INPUTS

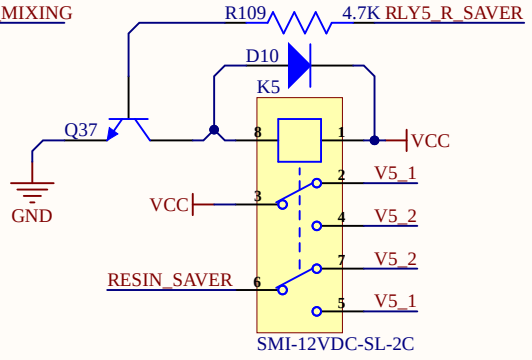


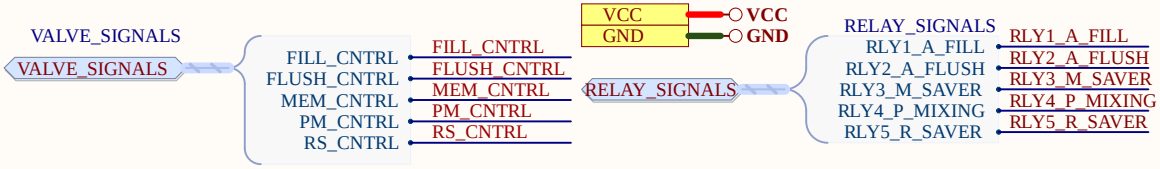




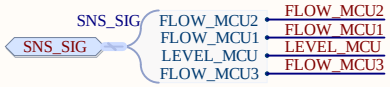




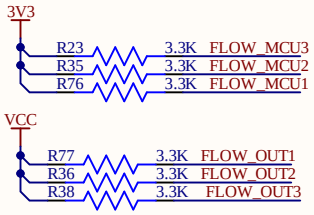




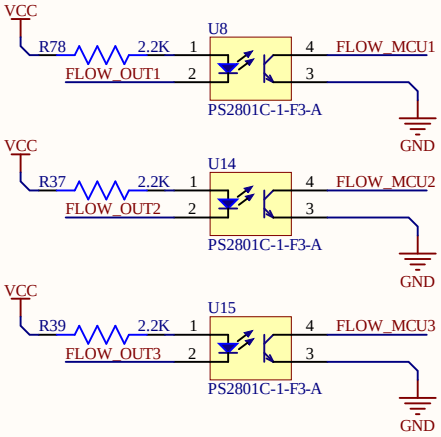
Title		
Size A	Number	Revision
Date: 2/23/2023	Sheet of	
File: E:\OliverStanton-WaterMonitor\..\Valve_Section.SchDoc	Drawn By:	



PULL UP RESISTORS

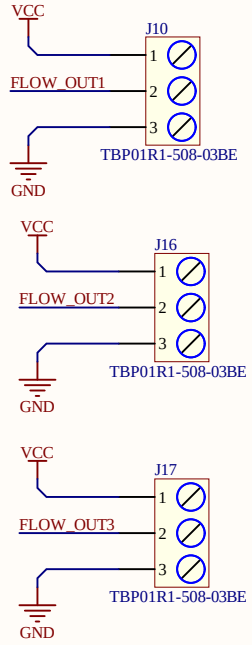


OPTO ISOLATION FOR FLOW METER



Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

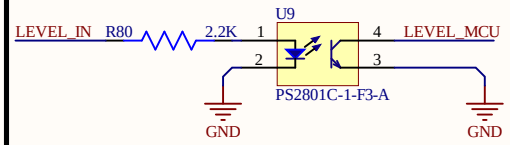
FLOW METER INPUT



SUPPORTING CIRCUIT

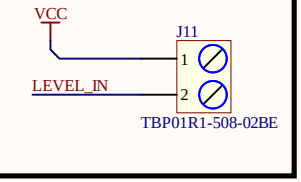


OPTO ISOLATION FOR FLOW METER



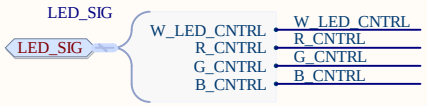
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

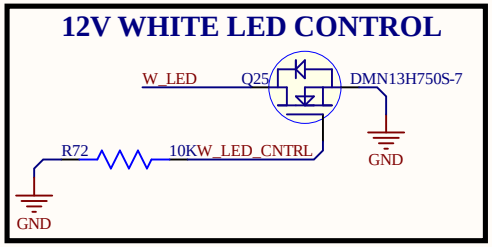
LEVEL SENSOR INPUT

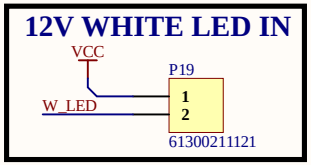


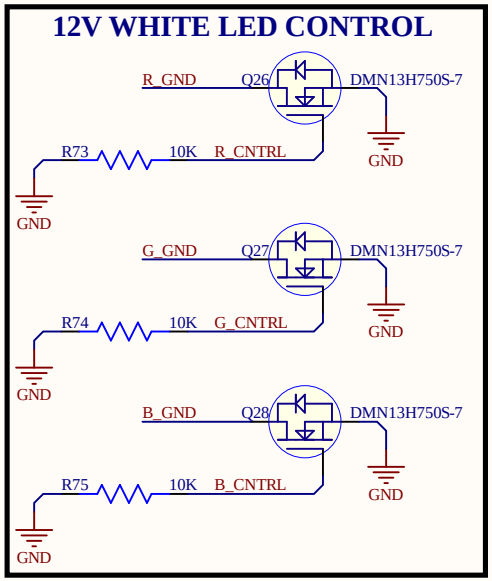


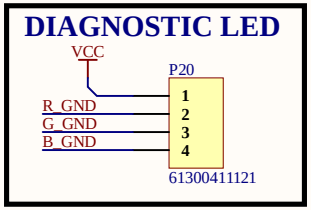
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Date: 2/23/2023		Sheet of
File: E:\OliverStanton-WaterMonitor\..\Sensor_Section.SchDoc		Drawn By:

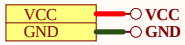




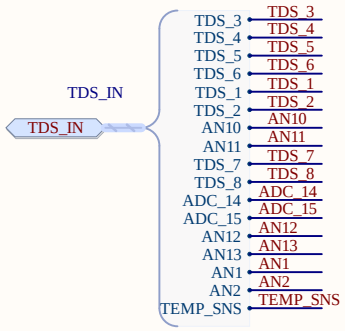




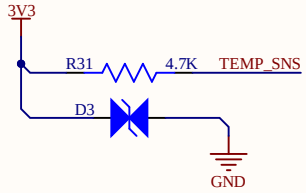




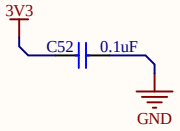
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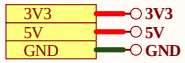


PULL UP RESISTORS

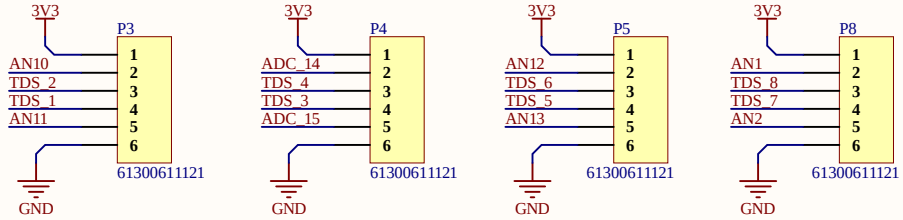


DECOUPLING CAP



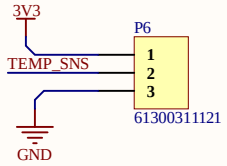


4 X TDS INPUTS

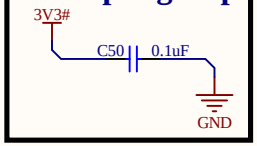


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

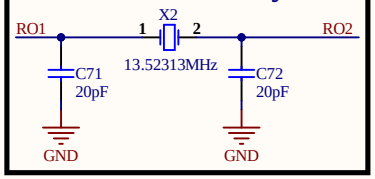
TEMP SENSOR DS18B20



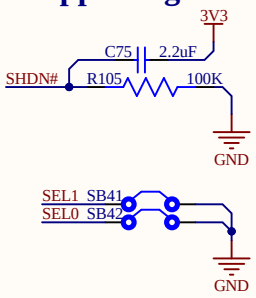
Decoupling Cap

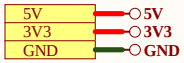


13.52127MHz Crystal



Supporting Circuit

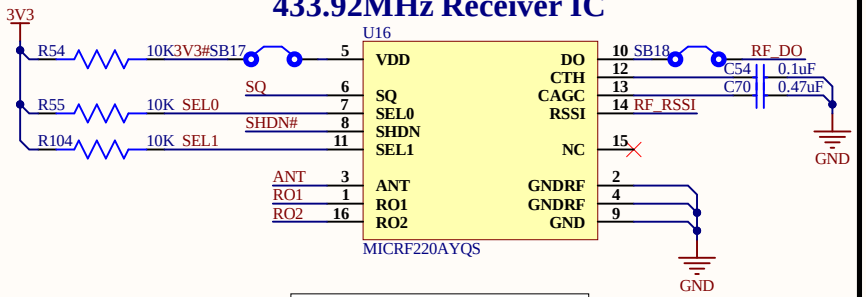






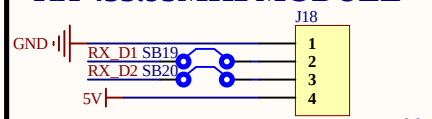


433.92MHz Receiver IC

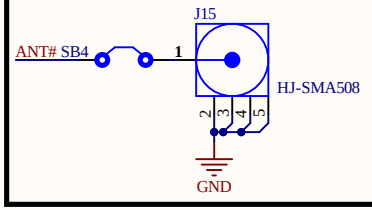


When SEL0 & SEL1 are pulled up, low pass filter BW = 13000Hz
Max encoded bit rate =20Kbps
Pulse width @433.92MHz=38us
CAGC Valu= 0.47uF
FREF= FRF/(32+87/1000)
FREF= 13.52323MHz (@433.92MHz FRF)

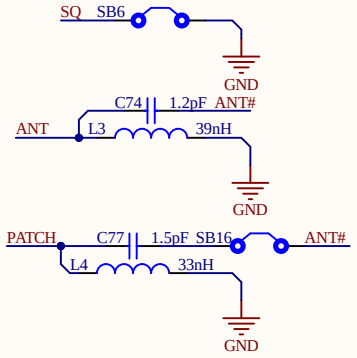
RX 433.93MHz MODULE



SMA CONNECTOR

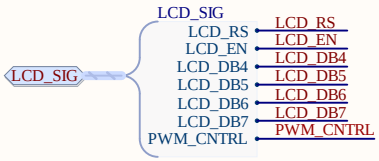


Z-MATCHING CIRCUITRY



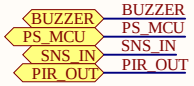
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Title: V3_Board.PrjPcb		
Size: A4	Number: 1	Revision: 1
Date: 2/23/2023	Sheet 16 of 17	
File: E:\OliverStanton-WaterMonitor\..\Receiver Section\Schdoc	Drawn By: Fuzaifa Ali	

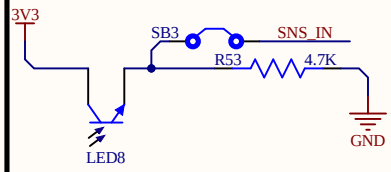






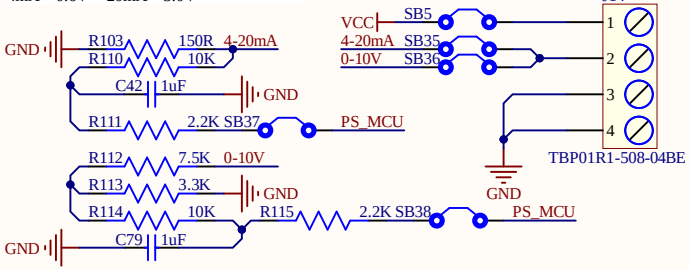


PHOTOTRANSISTOR

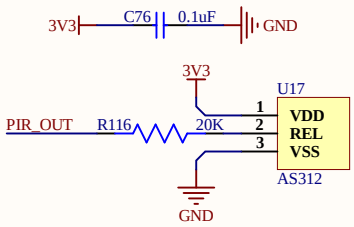


PRESSURE SENSOR

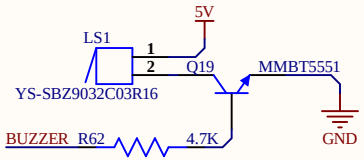
For 4-20mA respected ADC voltage will be:
4mA = 0.6V -- 20mA = 3.0V



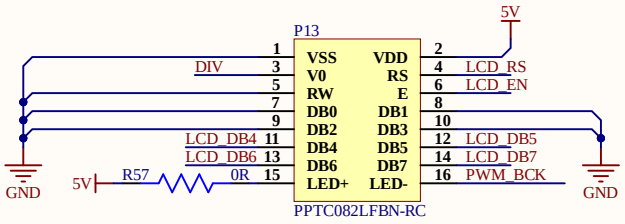
MOTION DETCTION



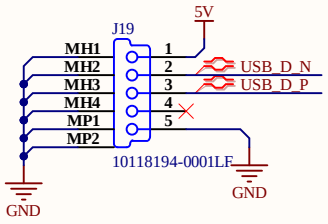
BUZZER CIRCUIT

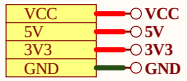


LCD CONNECTOR

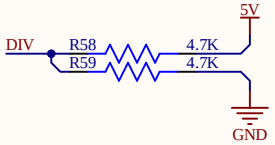


USB TYPE B CONNECTOR

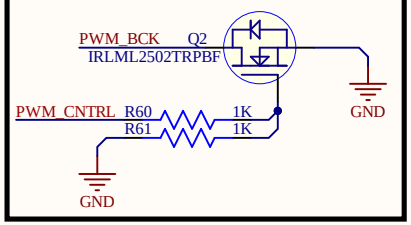




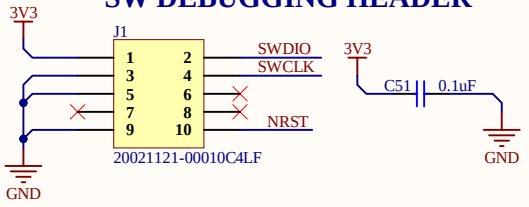
RESISTOR DIVIDER

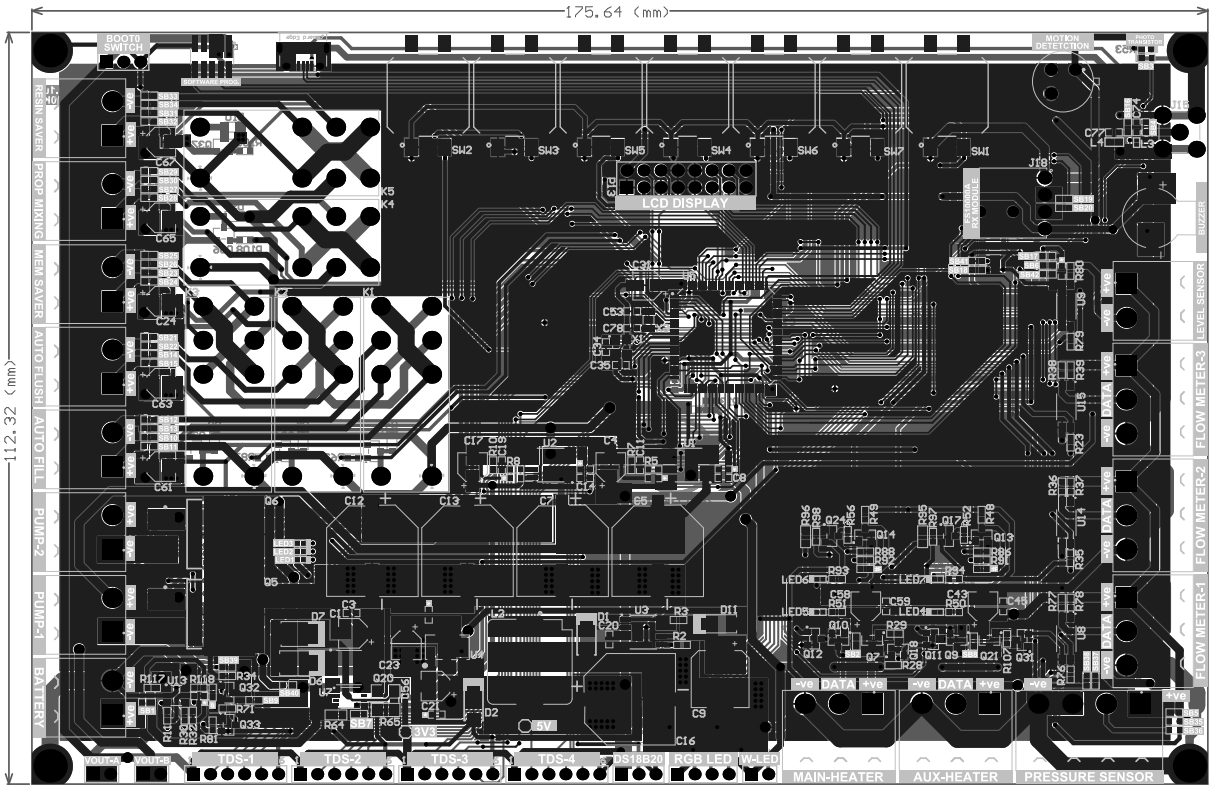


PWM BACKLIGHT



SW DEBUGGING HEADER





Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				

Note:
J18 is the header placed for FS10000A Receiver Module and it should be mounted on board with an offset of atleast 3mm from the surface of PCB.